Integrated Science 1 (#2002400)

Version: 2024 - And Beyond (current)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- SC.912.E.5.1: Cite evidence used to develop and verify the scientific theory of the Big Bang (also known as the Big Bang Theory) of the origin of the universe.
- SC.912.E.5.2: Identify patterns in the organization and distribution of matter in the universe and the forces that determine them.
- SC.912.E.5.4: Explain the physical properties of the Sun and its dynamic nature and connect them to conditions and events on Earth.
- SC.912.E.5.7: Relate the history of and explain the justification for future space exploration and continuing technology development.
- SC.912.E.5.8: Connect the concepts of radiation and the electromagnetic spectrum to the use of historical and newly-developed observational tools.
- SC.912.E.6.1: Describe and differentiate the layers of Earth and the interactions among them.
- SC.912.E.6.2: Connect surface features to surface processes that are responsible for their formation.
- SC.912.E.6.3: Analyze the scientific theory of plate tectonics and identify related major processes and features as a result of moving plates.
- SC.912.E.7.1: Analyze the movement of matter and energy through the different biogeochemical cycles, including water and carbon.

- SC.912.E.7.3: Differentiate and describe the various interactions among Earth systems, including: atmosphere, hydrosphere, cryosphere, geosphere, and biosphere.
- SC.912.L.14.1: Describe the scientific theory of cells (cell theory) and relate the history of its discovery to the process of science.
- SC.912.L.14.2: Relate structure to function for the components of plant and animal cells. Explain the role of cell membranes as a highly selective barrier (passive and active transport).
- SC.912.L.14.3: Compare and contrast the general structures of plant and animal cells. Compare and contrast the general structures of prokaryotic and eukaryotic cells.
- SC.912.L.14.4: Compare and contrast structure and function of various types of microscopes.
- SC.912.L.14.7: Relate the structure of each of the major plant organs and tissues to physiological processes.
- SC.912.L.15.1: Explain how the scientific theory of evolution is supported by the fossil record, comparative anatomy, comparative embryology, biogeography, molecular biology, and observed evolutionary change.
- SC.912.L.15.4: Describe how and why organisms are hierarchically classified and based on evolutionary relationships.
- SC.912.L.15.5: Explain the reasons for changes in how organisms are classified.
- SC.912.L.15.6: Discuss distinguishing characteristics of the domains and kingdoms of living organisms.
- SC.912.L.15.8: Describe the scientific explanations of the origin of life on Earth.
- SC.912.L.16.1: Use Mendel's laws of segregation and independent assortment to analyze patterns of inheritance.
- SC.912.L.16.14: Describe the cell cycle, including the process of mitosis. Explain the role of mitosis in the formation of new cells and its importance in maintaining chromosome number during asexual reproduction.
- SC.912.L.16.16: Describe the process of meiosis, including independent assortment and crossing over. Explain how reduction division results in the formation of haploid gametes or spores.
- SC.912.L.16.17: Compare and contrast mitosis and meiosis and relate to the processes of sexual and asexual reproduction and their consequences for genetic variation.

- SC.912.L.17.2: Explain the general distribution of life in aquatic systems as a function of chemistry, geography, light, depth, salinity, and temperature.
- SC.912.L.17.3: Discuss how various oceanic and freshwater processes, such as currents, tides, and waves, affect the abundance of aquatic organisms.
- SC.912.L.17.4: Describe changes in ecosystems resulting from seasonal variations, climate change and succession.
- SC.912.L.17.9: Use a food web to identify and distinguish producers, consumers, and decomposers. Explain the pathway of energy transfer through trophic levels and the reduction of available energy at successive trophic levels.
- SC.912.L.17.11: Evaluate the costs and benefits of renewable and nonrenewable resources, such as water, energy, fossil fuels, wildlife, and forests.
- SC.912.L.18.1: Describe the basic molecular structures and primary functions of the four major categories of biological macromolecules.
- SC.912.L.18.7: Identify the reactants, products, and basic functions of photosynthesis.
- SC.912.L.18.8: Identify the reactants, products, and basic functions of aerobic and anaerobic cellular respiration.
- SC.912.L.18.9: Explain the interrelated nature of photosynthesis and cellular respiration.
- SC.912.L.18.12: Discuss the special properties of water that contribute to Earth's suitability as an environment for life: cohesive behavior, ability to moderate temperature, expansion upon freezing, and versatility as a solvent.
- SC.912.N.1.1: Define a problem based on a specific body of knowledge, for example: biology, chemistry, physics, and earth/space science, and do the following:
    1. **Pose questions about the natural world,** (Articulate the purpose of the investigation and identify the relevant scientific concepts).
    2. **Conduct systematic observations,** (Write procedures that are clear and replicable. Identify observables and examine relationships between test (independent) variable and outcome (dependent) variable. Employ appropriate methods for accurate and consistent observations; conduct and record measurements at appropriate levels of precision. Follow safety guidelines).
    3. **Examine books and other sources of information to see what is already known,**
    4. **Review what is known in light of empirical evidence,** (Examine whether available empirical evidence can be interpreted in

terms of existing knowledge and models, and if not, modify or develop new models).

5. **Plan investigations,** (Design and evaluate a scientific investigation).
6. **Use tools to gather, analyze, and interpret data (this includes the use of measurement in metric and other systems, and also the generation and interpretation of graphical representations of data, including data tables and graphs),** (Collect data or evidence in an organized way. Properly use instruments, equipment, and materials (e.g., scales, probeware, meter sticks, microscopes, computers) including set-up, calibration, technique, maintenance, and storage).
7. **Pose answers, explanations, or descriptions of events,**
8. **Generate explanations that explicate or describe natural phenomena (inferences),**
9. **Use appropriate evidence and reasoning to justify these explanations to others,**
10. **Communicate results of scientific investigations, and**
11. **Evaluate the merits of the explanations produced by others.**

- SC.912.N.1.2: Describe and explain what characterizes science and its methods.
- SC.912.N.1.3: Recognize that the strength or usefulness of a scientific claim is evaluated through scientific argumentation, which depends on critical and logical thinking, and the active consideration of alternative scientific explanations to explain the data presented.
- SC.912.N.1.4: Identify sources of information and assess their reliability according to the strict standards of scientific investigation.
- SC.912.N.1.6: Describe how scientific inferences are drawn from scientific observations and provide examples from the content being studied.
- SC.912.N.1.7: Recognize the role of creativity in constructing scientific questions, methods and explanations.
- SC.912.N.2.1: Identify what is science, what clearly is not science, and what superficially resembles science (but fails to meet the criteria for science).
- SC.912.N.3.1: Explain that a scientific theory is the culmination of many scientific investigations drawing together all the current evidence concerning a substantial range of phenomena; thus, a scientific theory represents the most powerful explanation scientists have to offer.

- SC.912.N.3.3: Explain that scientific laws are descriptions of specific relationships under given conditions in nature, but do not offer explanations for those relationships.
- SC.912.N.3.4: Recognize that theories do not become laws, nor do laws become theories; theories are well supported explanations and laws are well supported descriptions.
- SC.912.N.3.5: Describe the function of models in science, and identify the wide range of models used in science.
- SC.912.P.8.1: Differentiate among the four states of matter.
- SC.912.P.8.2: Differentiate between physical and chemical properties and physical and chemical changes of matter.
- SC.912.P.8.3: Explore the scientific theory of atoms (also known as atomic theory) by describing changes in the atomic model over time and why those changes were necessitated by experimental evidence.
- SC.912.P.8.4: Explore the scientific theory of atoms (also known as atomic theory) by describing the structure of atoms in terms of protons, neutrons and electrons, and differentiate among these particles in terms of their mass, electrical charges and locations within the atom.
- SC.912.P.8.5: Relate properties of atoms and their position in the periodic table to the arrangement of their electrons.
- SC.912.P.8.7: Interpret formula representations of molecules and compounds in terms of composition and structure.
- SC.912.P.10.1: Differentiate among the various forms of energy and recognize that they can be transformed from one form to others.
- SC.912.P.10.4: Describe heat as the energy transferred by convection, conduction, and radiation, and explain the connection of heat to change in temperature or states of matter.
- SC.912.P.10.7: Distinguish between endothermic and exothermic chemical processes.
- SC.912.P.10.20: Describe the measurable properties of waves and explain the relationships among them and how these properties change when the wave moves from one medium to another.
- SC.912.P.12.3: Interpret and apply Newton's three laws of motion.
- MA.K12.MTR.1.1: **Actively participate in effortful learning both individually and collectively.**

  Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:
Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

- MA.K12.MTR.2.1: **Demonstrate understanding by representing problems in multiple ways.**

  Mathematicians who demonstrate understanding by representing problems in multiple ways:

  - Build understanding through modeling and using manipulatives.
  - Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
  - Progress from modeling problems with objects and drawings to using algorithms and equations.
  - Express connections between concepts and representations.
  - Choose a representation based on the given context or purpose.

Clarifications:
Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.
- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.

- MA.K12.MTR.3.1: **Complete tasks with mathematical fluency.**

  Mathematicians who complete tasks with mathematical fluency:

  - Select efficient and appropriate methods for solving problems within the given context.
  - Maintain flexibility and accuracy while performing procedures and mental calculations.
  - Complete tasks accurately and with confidence.
  - Adapt procedures to apply them to a new context.
  - Use feedback to improve efficiency when performing calculations.

  Clarifications:
  Teachers who encourage students to complete tasks with mathematical fluency:

  - Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
  - Offer multiple opportunities for students to practice efficient and generalizable methods.
  - Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- MA.K12.MTR.4.1: **Engage in discussions that reflect on the mathematical thinking of self and others.**

  Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

  - Communicate mathematical ideas, vocabulary and methods effectively.
  - Analyze the mathematical thinking of others.

- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.
- Construct possible arguments based on evidence.

Clarifications:
Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

- MA.K12.MTR.5.1: **Use patterns and structure to help understand and connect mathematical concepts.**

  Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

  - Focus on relevant details within a problem.
  - Create plans and procedures to logically order events, steps or ideas to solve problems.
  - Decompose a complex problem into manageable parts.
  - Relate previously learned concepts to new concepts.
  - Look for similarities among problems.
  - Connect solutions of problems to more complicated large-scale situations.

  Clarifications:
  Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

  - Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
  - Support students to develop generalizations based on the similarities found among problems.

    - Provide opportunities for students to create plans and procedures to solve problems.
    - Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- MA.K12.MTR.6.1: **Assess the reasonableness of solutions.**

  Mathematicians who assess the reasonableness of solutions:

    - Estimate to discover possible solutions.
    - Use benchmark quantities to determine if a solution makes sense.
    - Check calculations when solving problems.
    - Verify possible solutions by explaining the methods used.
    - Evaluate results based on the given context.

  Clarifications:
  Teachers who encourage students to assess the reasonableness of solutions:

    - Have students estimate or predict solutions prior to solving.
    - Prompt students to continually ask, "Does this solution make sense? How do you know?"
    - Reinforce that students check their work as they progress within and after a task.
    - Strengthen students' ability to verify solutions through justifications.

- MA.K12.MTR.7.1: **Apply mathematics to real-world contexts.**

  Mathematicians who apply mathematics to real-world contexts:

    - Connect mathematical concepts to everyday experiences.
    - Use models and methods to understand, represent and solve problems.
    - Perform investigations to gather data or determine if a method is appropriate.
    - Redesign models and methods to improve accuracy or efficiency.

  Clarifications:
  Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
- Challenge students to question the accuracy of their models and methods.
- Support students as they validate conclusions by comparing them to the given situation.
- Indicate how various concepts can be applied to other disciplines.

- ELA.K12.EE.1.1: Cite evidence to explain and justify reasoning.

  Clarifications:
  K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

  2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

  4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

  6-8 Students continue with previous skills and use a style guide to create a proper citation.

  9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- ELA.K12.EE.2.1: Read and comprehend grade-level complex texts proficiently.

  Clarifications:
  See Text Complexity for grade-level complexity bands and a text complexity rubric.

- ELA.K12.EE.3.1: Make inferences to support comprehension.

  Clarifications:
  Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like "Why is the girl smiling?" or make predictions about what will

happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- ELA.K12.EE.4.1: Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

  Clarifications:
  In kindergarten, students learn to listen to one another respectfully.

  In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think ________ because _______.” The collaborative conversations are becoming academic conversations.

  In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- ELA.K12.EE.5.1: Use the accepted rules governing a specific format to create quality work.

  Clarifications:
  Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- ELA.K12.EE.6.1: Use appropriate voice and tone when speaking or writing.

  Clarifications:
  In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd grade and beyond, students practice appropriate social and academic language to discuss texts.

- ELD.K12.ELL.SC.1: English language learners communicate information, ideas and concepts necessary for academic success in the content area of Science.
- ELD.K12.ELL.SI.1: English language learners communicate for social and instructional purposes within the school setting.

Course Information